REAL-TIME ADAPTIVE LANE DETECTION FOR INDIAN ROADS

A Capstone Project report submitted
in partial fulfillment of requirement for the award of degree

BACHELOR OF TECHNOLOGY
in
SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE
By

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ABSTRACT

One of the most important technologies in advanced driver assistance systems and autonomous driving is lane detection, which enables a vehicle to go through the route and eliminate potential hazards. This project was designed to build a robust lane detection system for Indian roads because Indian roadways pose their own specific challenges such as faded or inconsistent lane markings, varied road conditions, diverse traffic patterns, and frequent obstructions. This system proposes to detect lane boundaries accurately and in real time using a combination of advanced image processing methods , computer vision tools and deep learning approaches such as OpenCV. Techniques like edge detection, Hough Transform and HSV color space for color based filtration are adapted to improve performance in diverse conditions. The project aims to optimize it for practical use on Indian roads, so it should be able to handle missing lane markings and irregular road layouts. Preliminary results show that considerable improvements in detection accuracy and robustness over conventional approaches are obtained. This project addresses the immediate need for safer and more reliable navigation tools but contributes, at large, to the adoption of ADAS and autonomous driving technologies in India, which can be a critical foundation for future intelligent transportation systems tailored for emerging markets.

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CHAPTER-1
INTRODUCTION

Lane detection is the basic step in navigation, lane departure warning, and path planning for advanced driver assistance and self-governing vehicles. While there are many developments globally towards lane detection technologies, the systems still face challenges to adapt to the Indian road conditions as they are unique and more diverse. Indian roads have faded lane markings with worn-out or non-existent lane signs, unpredictable behavior of the traffic, and road blockages such as parked vehicles, pedestrians, and animals. Environmental conditions are an added source of difficulty, with different lighting conditions, seasonal rains, and road deterioration that interfere with reliable lane detection.

This project develops a lane detection system for Indian roads by overcoming all these difficulties. The system utilizes state-of-the-art computer vision for real-time identification of lane boundaries, even under difficult conditions. Images go through preprocessing, which removes the noise, detects features using image segmentation techniques and edge detection techniques for accurate lane identification.

The significance of the project lies in overcoming most of the difficulties faced in lane detection on Indian roads, it adds up to the greater goal of developing intelligent transportation systems for raising markets as it creates a safer and more efficient drive.

1.1 EXISTING SYSTEM:

Although many methods like edge detection and the Hough Transform are available for the traditional techniques of image processing, their

performance decreases when the complexity in road conditions becomes very serious. Approaches through machine learning, consisting of algorithms like SVM, provide a degree of adaptability and flexibility. However, they are still limited concerning the manipulation of the address of numerous scenarios in real-world applications. Applications with deep learning-based approaches, including advanced U-Net and LaneNet, show significantly higher accuracies compared to conventional approaches. However, deep learning methods generally need large sets of training data and incur high computational costs to achieve their respective gains. Hybrid systems were proposed concentrating on combining the gains both in terms of efficiency as well as accuracy by providing the benefits of traditional approaches along with the advanced deep learning methods. These hybrid systems are known to work well on structured roads in commercial ADAS applications. However, they tend to face significant challenges and hardship when applied to Indian roads, characteristically less structured and more complex in nature. All these approaches, however, have crucial shortcomings in problems related to missing lane markings, high computational demand by advanced algorithms, and issues of adaptability by unstructured driving conditions.

1.2 PROPOSED SYSTEM:

The proposed lane detection system is particularly designed to address the challenging conditions commonly seen on Indian roads. These include inconsistent or faded lane markings which cause significant impairment to visibility and cluttered views from various vehicles, pedestrians, and the unpredictable changes in light and weather. Under these conditions, lane detection is a bit more complicated, requiring a highly flexible system that would adapt to these diverse conditions yet remain accurate in its detections.

Key Features

- Straight Lane Detection:

By applying the Canny operator and the Hough Transform for edge detection, lane boundaries can be highlighted effectively. Color-based detection using the HSV color space allows the system to adapt to variations in road surface and illumination, enhancing visibility and detection accuracy under diverse conditions.

- Real-Time Processing:

Optimized for high-quality lane detection and tracking in real-time, the algorithm is designed to minimize latency, enabling seamless execution in practical applications.

- Indian Road Adaptation:

The system is customized to handle challenges specific to Indian roads, such as faded, inconsistent, or partially corked lane markings; clutter on the roads; and variable weather conditions including poor lighting and rain.

- Kalman Filter and Exponential Moving Average (EMA):

A Kalman Filter is used to predict lane offsets accurately, even in noisy or cluttered environments, while the EMA smooths lane boundary estimations, contributing to system robustness and providing a stable lane-tracking experience.

CHAPTER-2
LITERATURE SURVEY

Subramani, R., Suresh, K., Cecil, D., & Vijayalakshmi, C. [1] This research paper primarily focuses on lane detection under different conditions which also includes different time scenarios like day and night due which the shadows created by the sun light and street lights could be a problem for detecting the lanes. The challenges faced in this study are poor lighting, different weather conditions and roads conditions. To solve these challenges they used preprocessing techniques and Kalman filter which have finally improved the overall accuracy of the project. The preprocessing techniques used in this paper are Gamma Correction, Region of interest, bilateral filtering, HSV filtering, Canny Edge detection and hough transform. R lane detection process clustering is used in this paper. Density Based Spatial Clustering of application with Noise is used for group relevant lanes. But the Limation of this paper is it doesn't work for curved lanes.

Kreucher, C., Lakshmanan, S., & Kluge, K. (1998, October) [2] This paper basically developed a system that gives an alert or intimate the driver about the lane departures with the help of LOIS algorithm for detecting lanes and Kalman filter for prediction of lanes. The Primary goal or Objective of this paper is to provide a strong driving support for the driving in particular situations like drowsiness or when driver is out of focus. The most prominent algorithm used in this paper is Lois algorithm. Lois stands for Likelihood of Image Shape. This algorithm works perfect even in conditions like shadows and varying lighting sources. At the starting of predictions the algorithm works weaker but with subsequent tracking, the algorithm uses information from previous frames which results in better accuracy and performance. This system created an intimation for the driver when the predicted position lies within the predefined threshold.

Dorj, B., Hossain, S., & Lee, D. J. (2020) [3] This Paper has implemented a system that detects curved lanes which are critical for autonomous or self-driving vehicles, specially on roads which have significant turns. It combines Kalman filter with geometric models which resulted to detect and track curved lanes with better accuracy. The primary objective of this paper is to improve the safety by predicting accurate curve road turns. The faced challenges are uncertainties in the image data like noise. They used Otsu's thresholding method and top-view image transformation for lane detection. In case of

straight line hough transform is used and in case of curve lanes, it uses Kalman filter with integration of further 2 models: parabolic model-used to fit the curve with parabolic equation and Circular model- used to fit the curve with circle equation with estimating its center and radius. This work mainly helps in adjusting the steering angle based upon the road's curvature.

Samyak Shah, A. J. (2022) [4] The paper presents a real time road lane detection system

boundaries, and warns drivers of unintended lane departures and can be used for autonomous vehicle technologies. The proposed solution in the paper refines the current methods like edge detection, region of interest (ROI) segmentation, and hough transform for better accuracy and efficiency. It uses image processing tools like grayscale conversion, gaussian blur for noise reduction, and hough transform for lane line detection. The system differentiates the lane markings through color masks and focuses on a defined ROI to minimize irrelevant edge detections. The Hardware integration in this paper includes a Raspberry Pi Zero with a camera to process images and deliver real time audio warnings. It demonstrates high accuracy, i.e., over 96% even in non- optimal weather conditions, improving

the road safety by preventing lane departure related accidents. The future scope includes improving curved lane detection, optimizing performance in extreme weather and incorporating databases for hazard recognition.

Rajakumar.Ra, 1. P. (2021) [5] The paper presents a real time road lane detection system which can be used for reducing traffic accidents caused by not attentive driving. The system detects lane boundaries, and warns drivers of unintended lane departures and can be used for autonomous vehicle technologies. The proposed solution in the paper refines the current methods like edge detection, region of interest (ROI) segmentation, and hough transform for better accuracy and efficiency. It uses image processing tools like grayscale conversion, gaussian blur for noise reduction, and hough transform for lane line detection. The system differentiates the lane markings through color masks and focuses on a defined ROI to minimize irrelevant edge detections. The Hardware integration in this paper includes a Raspberry Pi Zero with a camera to process images and deliver real time audio warnings. It demonstrates high accuracy i.e over 96% even in non optimal weather conditions, improving the road safety by preventing lane departure related accidents. The future scope includes improving curved lane detection, optimizing performance in extreme weather and incorporating databases for hazard recognition.

Shipra Singh, S. M. (2024, May) [6] The paper titled presents a system for curved lane detection, which is crucial for vehicles and advanced driver systems (ADAS), to enhance road safety and navigation. In the paper the methodology includes combining traditional image processing with advanced algorithms, overcoming challenges such as inconsistent lane markings, lighting variations, and complex road geometries. Images are preprocessed through grayscale conversion, noise reduction, thresholding, perspective transformation for bird's eye view, and Sobel edge detection to highlight lane

edges. Here, the detection process uses a sliding window technique to travel lanes and a quadratic curve fitting algorithm to represent lane curvature, which calculates the radius of curvature for precise results. The testing gives high accuracy, with robust performance across different conditions, reducing errors like false positives and lane deviations. The areas are improved as the system adapts to various environments like urban roads and mountainous terrains.

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The author mentioned future improvements include handling extreme weather conditions, enabling vehicle-to-vehicle communication for collective road awareness, and extending capabilities to detect pedestrian lanes and crosswalks.

Kaur, G., & Kumar, D. (2015) [7] This paper provides a brief outline of various lane detection methods which are utilized in intelligent transportation systems to improve road safety and reduce accidents. The main objective of this paper says that lane detection plays an important role in self- driving cars. This paper emphasis mainly about improving lane detection algorithms under various conditions. The existing algorithms works perfect for ideal conditions but struggles with real-world scenarios. The algorithms used in this paper is hough transform canny edge detection and bilateral filtering. Basically, this paper proposes a mixture of filtering and detection techniques to overcome the challenges.

Goel, A. (2014) [8] This paper performs different methodologies for detection of lanes in intelligent transportation systems which aims to overall improving of road safety by preventing lane deviations. Algorithms used are hough transform and canny edge detection which are effective under ideal roads but faces difficulties in real world scenarios. The advanced approaches integrates multiple algorithms with geometric models offer better accuracy. This paper identifies gaps in existing systems particularly in cases the algorithm prefers high quality images for accurate results and inability to handle extreme conditions and suggests future improvements on the algorithms for different roads and whether conditions.

Jingwei Cao, C. S. (2019) [9] The paper presents a lane detection algorithm for intelligent

vehicles that are operated in complex and critical environments. The algorithm overcomes the limitations of traditional and deep learning methods by achieving. high accuracy and real time performance. It starts by correcting image distortions using camera calibration and inverse perspective transformation to provide an aerial view of the lane or view from above. The edge detection is enhanced by combining the Sobel operator and HSL color space thresholding. Lane lines are fitted using RANSAC algorithm which is based on a third order Bspline curve model, enabling strong detection even in challenging conditions. Experimental validation on road driving videos and the Tusimple dataset showed an average detection accuracy of 98.49% and 98.42%, with average processing times of approximately 21.5 ms per frame. The algorithm outclassed traditional methods and some deep-learning approaches, demonstrating excellent anti-interference, adaptability, and efficiency. It was tested under various conditions such as highways, mountain roads, and tunnels, maintaining reliable performance despite challenges like lighting, and road curvature. The study concludes that the

proposed algorithm significantly enhances vehicles' safety and technical capabilities of vehicles. The future work includes aiming to improve inclusiveness and error detection to further optimize the algorithm's overall performance.

Arindol Das, [. S. (2024) [10] The paper presents a lane and curve detection system using deep learning to improve safety and for reducing accidents. It incorporated traditional computer vision techniques with convolutional neural networks (CNN) for robust lane detection, particularly for curved lanes. The process starts with data preprocessing which involves gradient calculation using Sobel and Laplacian operators, Gaussian smoothing for noise reduction, and Canny edge detection for identifying lane boundaries. After preprocessing the detected lane boundaries are mapped onto original road images for visualization. The CNN model, a Fully Convolutional Network (FCN), is used to create a binary segmentation map for lane pixels. The model undergoes training with a different types of annotated datasets, and its performance is fine-tuned using various hyperparameter optimization techniques. The approach combines the accuracy of deep learning with the speed and simplicity of traditional methods. The testing on different things demonstrates its capability to adapt to various road conditions, achieving precise detection of lane markings and curvature. In Future work, further model optimizations to enhance performance and expand usability.

CHAPTER-3

DESIGN

A lane detection system for Indian roads typically involves following steps:

- **Data Preprocessing:** Frames from video input are processed, making them clear and uniform by resizing, normalization, improving the quality (contrast adjustment), and reducing noise. For Indian roads, the quality of images declines due to clutter and environmental conditions.
- **Model Architecture:** It uses a hybrid approach on traditional image processing Canny edge detection, Hough Transform- as well as HSV color-based segmentation towards the lane-marking detection. Kalman filter predicts offsets with high accuracy and EMA smooths the lane boundaries for quick tracking.
- **Evaluation:** Lane Detection Consistency, Offset Stability, FPS evaluate its reliability in detecting lanes on cluttered Indian roads.

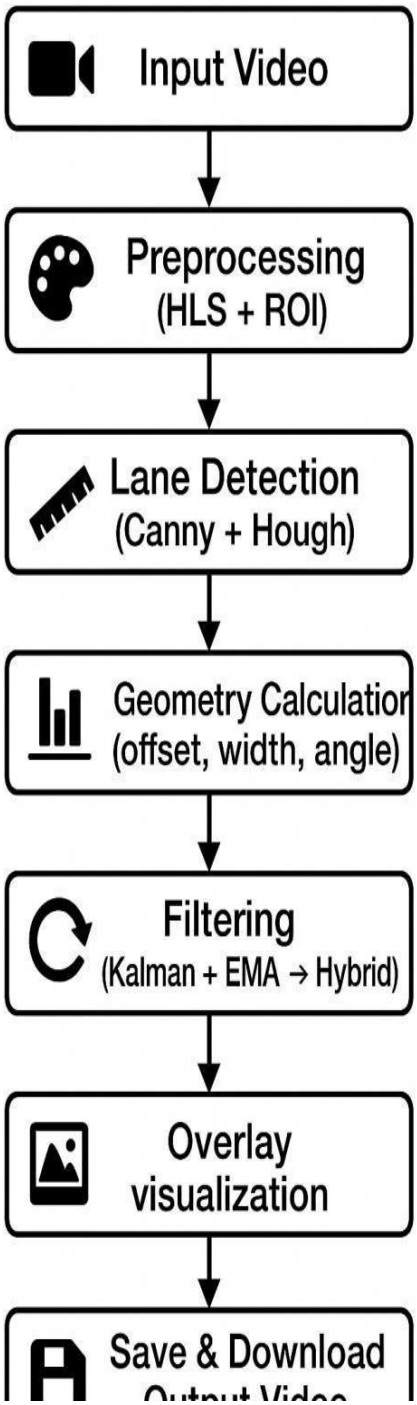


Fig 3.1 Steps in Lane Detection System (Straight)

3.1 REQUIREMENT SPECIFICATION

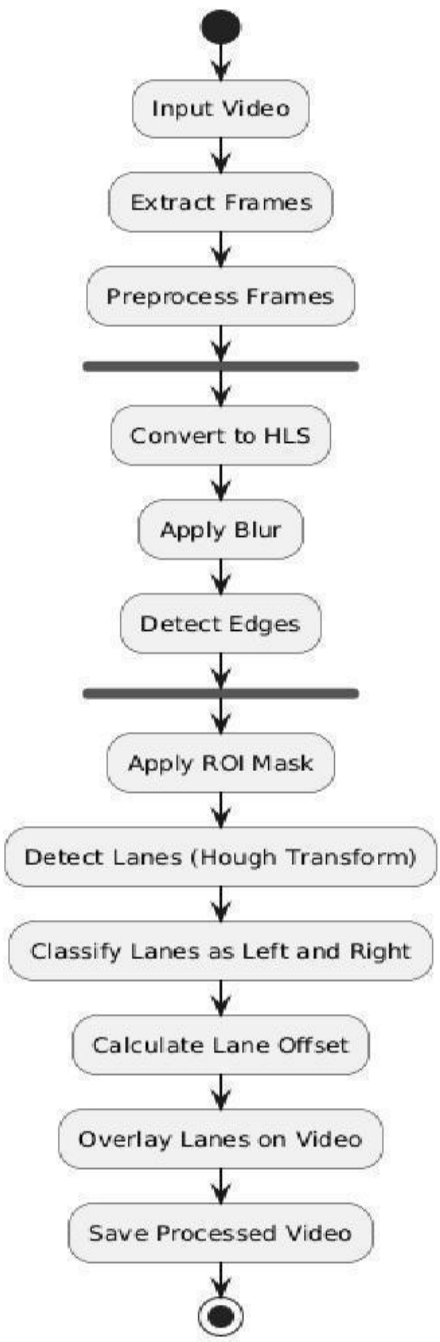
3.1.1 SOFTWARE REQUIREMENTS:

- Programming Language: Python (preferred), or any language that supports machine learning libraries (TensorFlow, OpenCV).
- Image Processing Libraries: OpenCV for resizing, edge detection, and color space transforms.
- Math Library: Using NumPy for the numerical operations involved in changing images and filter implementations.
- Optional: Visualization libraries like Matplotlib for plotting results of detections and metrics performance.

3.1.2 HARDWARE REQUIREMENTS:

- Processor (CPU): Multi-core processor (i5 or equivalent) for smooth operation and real-time processing.
- Memory (RAM): Minimum 8 GB, recommended; more video file processing performance enhances with more RAM.
- Storage: Storage of the processed data.
- Graphics Processing Unit: Though this component is optional, having a GPU, such as NVIDIA GTX or AMD Radeon, will significantly speed up real-time processing, especially when considering deep learning.

3.2 UML DIAGRAM:



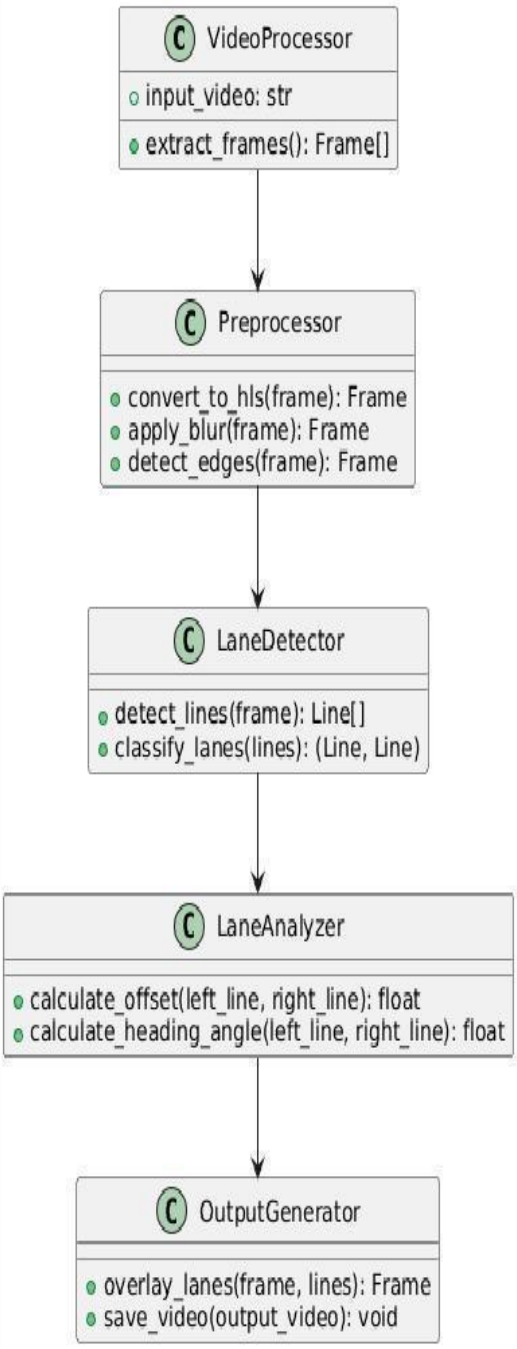


Fig 3.3 Activity Diagram Fig 3.4 Class Diagram

CHAPTER-4
IMPLEMENTATION

4.1 Preprocessing :

Preprocessing is one of the foundational steps in any computer-vision–based lane detection pipeline. Raw frames captured from an onboard vehicle camera are often influenced by multiple degrading factors such as motion blur, shadows, fluctuating environmental illumination, asphalt texture variations, worn-out lane markings, vehicle glare, and camera noise. If these imperfections are not corrected early, they propagate to later stages, causing inaccurate edge detection, false positives, and unstable lane tracking.

The primary objective of preprocessing is to standardize, clean, and enhance each video frame so that reliable lane features can be extracted. Proper preprocessing improves robustness across daytime, nighttime, rainy weather, and uneven road conditions.

- **Resizing the Frames :**

A fixed resolution is applied to all incoming video frames. This accomplishes two goals:

Uniformity: This is necessary for consistent mathematical operations since it guarantees that each frame processed by the subsequent stages has the same size.

Performance Improvement: By reducing the computing load, a smaller resolution enables the system to run at a higher frames-per-second (FPS) in real time.

To balance accuracy and processing speed, the majority of real-time systems scale frames to 640x480 or comparable resolutions.

- **Contrast Enhancement using HLS Color Space :**

Lighting affects road lane markings, particularly white and yellow lines. To successfully draw attention to these markings:

i. Frame Conversion: RGB/BGR color space is transformed to HLS (Hue, Lightness, Saturation) for each frame.
   a. Bright lane lines can be identified with the aid of the Lightness channel.
   b. Shadows and background are reduced with the aid of the saturation channel.

ii. Binary Masking: To separate lane markers from the surrounding area, a mask is used to remove only the white areas of the frame.

This aids in eliminating trees, shadows, cars, background objects, and asphalt texture.

iii. Contrast Boosting: It is simpler to identify fading, low-visibility, or worn-out lane markers when the lightness region is improved.

This phase greatly enhances the future edge detection algorithm's performance.

- **Noise Reduction using Gaussian Blur :**

Even after applying the binary mask, the result may still contain undesired edges, harsh gradients, or tiny dots because of:

i. texture of the road,
ii. cracks,
iii. shadows,
iv. noise from the camera.

Gaussian Blur is used to smooth out these abnormalities.

Gaussian Blur: Why Use It?

High-frequency noise is eliminated.

It creates smoother gradients and softens the edges.

During Canny edge extraction, it guarantees that only significant edges (lane lines) show up.

As a result, lane boundaries are prominently displayed in a clear, noise-reduced picture.

Edge Detection:

The system, then employs algorithms for edge detection, i.e., Canny edge detector to aid in the search for edges that might be helpful to detect lane boundaries. The line found is then considered a straight lane line from an edge-detected image using the Hough Transform function.

Fig 4.1 Input image

Fig 4.2 HLS color space

Fig 4.3 Mask white and Gaussian blur

Fig 4.4 Canny edge detection

Filtering and smoothing:

The system employs a combination of Kalman Filtering and Exponential Moving Average (EMA) to achieve highly stable and smooth lane tracking across video frames.

- Kalman Filter for Predictive Lane Stabilization :

The Kalman Filter plays a vital role in stabilizing the detected lane boundaries. Real-world road environments contain many disturbances—such as shadows, cracked pavements, uneven brightness, sudden steering movement, and temporarily missing lane markings—which can cause fluctuations in lane detection. The Kalman Filter overcomes these issues by:

Using Past Observations: It analyzes lane positions detected in previous frames to build a mathematical prediction model.

Predicting Future Lane Positions: When lane visibility is poor or partially missing (for example, in broken or faded markings), the filter predicts the most likely lane position rather than relying solely on the current noisy frame.

Correcting Abrupt Jumps: If there is a sudden shift in detection due to noise, the filter smoothens the transition and avoids jerky movements in the final output.

Reducing Detection Noise: It suppresses random fluctuations and ensures the detected lane lines remain stable throughout the video sequence.

- Exponential Moving Average (EMA) for Real-Time Lane Tracking:

While the Kalman Filter provides predictive smoothing, the Exponential Moving Average (EMA) adds an additional layer of real-time stability. EMA works by giving more weight to recent lane detections while still considering previous

frames. Its benefits include:

Smoother Transitions: It prevents lane lines from constantly shifting left and right when the algorithm detects minor edge variations.

Better Performance on Curved Roads: Due to its adaptive nature, EMA responds smoothly to gradual changes in lane curvature.

Continuous Real-Time Tracking: Even if edges fluctuate slightly, EMA maintains a visually stable lane line, making the final output appear clean and consistent.

Together, Kalman + EMA create a dual-layer smoothing system, ensuring the lane detection remains highly stable, visually smooth, and robust even under challenging conditions.

Evaluation :

To validate system performance, several key parameters are calculated every 10 frames, allowing for consistent monitoring of accuracy and stability.

Lane Width, Heading Angle, and Offset Analysis

The evaluation includes:

- Lane Width: Ensures the system accurately measures the distance between left and right lane boundaries.
- Heading Angle: Shows whether the vehicle is aligned with the lane or drifting.
- Offset: Indicates how far the vehicle is from the center of the lane.

These metrics help verify both detection accuracy and system reliability in real-world scenarios.

Lane Detection Consistency – 99.91%

The system maintains 99.91% consistency, meaning it almost always identifies lane lines correctly across various driving conditions such as:

- curved or straight roads,
- low-contrast surfaces,
- illumination changes,
- partial lane markings.

This high consistency demonstrates the robustness of the preprocessing, edge detection, and filtering pipeline.

Offset Estimation – Average 0.36

An offset value of 0.36 indicates that the system can reliably estimate how much the vehicle is deviating from the lane center. Low offset variation shows:

- stable lane center detection,
- highly accurate interpretation of vehicle positioning,
- suitability for driver-assistance and autonomous driving applications.

Real-Time Processing Speed – 83.32 FPS

The system achieves 83.32 Frames Per Second (FPS) on average, which is significantly higher than the standard video rate. High FPS ensures:

- minimal delay in lane overlay,
- responsive adjustments during curves or sharp turns,
- microsecond-level stability improvements due to rapid frame processing.

Such speed makes the system practical for real-time driving scenarios.

Robustness to Road Variations

The system remains strong under multiple conditions:

- Faded or broken lane markings,
- Road clutter (vehicles, shadows, roadside objects),
- Different road textures, including asphalt, concrete, and patchwork surfaces,
- Environmental variations, such as bright sunlight, overcast lighting, and mild shadows.

This level of robustness confirms that the lane detection method can adapt to challenging real-world conditions without compromising accuracy.

4.1 MODULES

Core Libraries for Image Processing:

- **OpenCV:** Used for image preprocessing (resizing, edge detection) and lane detection tasks such as Canny edge detection and Hough Transform.
- **NumPy:** Supports Performing Numerical Operations Related to

Image Transformations and Matrix Computation.

Data Handling and Augmentation:

- **NumPy:** To manipulate and process the road image dataset.
- **Augmentation libraries:** To mimic diverse conditions on the road, robustly improving the model.

Data Visualization:

- **OpenCV (Real-time Visualization):** This is used to superimpose detected lanes over the video frames during real-time processing.

Lane Detection and Tracking Algorithm:

- **Kalman Filter:** Determines future lane positions; thus, no positional inaccuracies.
- **Exponential Moving Average (EMA):** This helps smooth the lane boundary and ensures stable and consistent lane tracking.

4.2 OVERVIEW TECHNOLOGY:

The Challenge:

Lane detection in Indian roads will pose a very challenging task as the road markings vary and the traffic density is heavy, etc. Thus, all these issues do cause problems in detecting lanes consistently in real-world or unstructured environments.

Why Hybrid Models Excel:

This kind of problem highly suited to a hybrid approach combining traditional image processing-such as edge detection and Hough Transform-with some real-time tracking algorithms like Kalman Filter and EMA. The

proposed method is very efficient in the detection of lane boundaries, even with road noises and variable lighting, while ensuring real-time processing.

How the Model Works:

- **Input**: The system takes in road images or video frames.
- **Edge detection**: Canny edge detection and the Hough Transform are used in algorithms for detecting lane boundaries using techniques in image processing. Color- based segmentation is also used to outline lane markings under various lighting conditions.
- **Optimization:** The Kalman Filter serves to forecast lane positions across temporal intervals, whereas the Exponential Moving Average (EMA) facilitates the maintenance of smooth lane tracking, despite varying environmental conditions.
- **Lane Detection:** Image processing combined with tracking gives the accurate lane detection as well as real time position estimations of the lanes.

Benefits:

- **High Accuracy:** The hybrid model can obtain very high accuracy in lane detection with difficult roadway conditions such as debris, worn lane markings, and inadequate illumination. The system can process real-time video frames, so it is very feasible to use in autonomous vehicle applications.
- **Robustness:** The model withstands road noise, changes in traffic conditions, and styles of lane markings on Indian roads effectively.

  Below tables shows the results obtained by processing the same input for Kalman optimization, EMA and hybrid approach.

Optimization	*Lane Detection Consistency*	*Offset stability*	*FPS*
Kalman Filter	99.52%	16.69 px	69.68
Exponential Moving Average (EMA)	99.52%	39.74 px	81.88
Hybrid Offset (Kalman + EMA)	99.88%	0.53 px	79.94

Table 4.1 Comparison of different methods

The hybrid method outperforms both Kalman and EMA in lane detection consistency and offset stability, while maintain a high FPS. This combination could be potential approach for real-time lane detection with good accuracy and performance speed.

Materials for Further Reading:

- Hybrid models for lane detection methodologies can provide interesting insights from scholarly articles on what may come next.
- Tutorials on the usage of OpenCV and some Python libraries like NumPy and Matplotlib can implement and visualize lane detection systems.

4.3. OPTIMIZATION TECHNIQUES FOR LANE DETECTION:

This study applies several optimization strategies that improve lane tracking consistency, lower noise, and provide smooth offset estimates in order to guarantee steady and accurate lane recognition under dynamic Indian road circumstances. The Kalman Filter, the Exponential Moving Average (EMA), and a hybrid optimization

model that combines both approaches are the main optimization strategies used.

4.3.1 Kalman Filter-Based Prediction :

The vehicle's lane deviation is predicted and corrected in real time using the Kalman Filter, a state estimation approach. The sequential processing of video frames can result in unpredictable outcomes due to small deviations or missing lane detections.

This is lessened by the Kalman Filter by:

- Predicting the next lane position using previous state information.
- Correcting the prediction based on the new measurement from the current frame.
- Reducing the influence of noise caused by lighting, shadows, or missing lane lines.

4.3.2 Exponential Moving Average (EMA) Smoothing :

To lessen sudden changes, the observed offset values are subjected to the Exponential Moving Average (EMA) approach. A temporal smoothing technique offered by EMA gradually fades older data while giving priority to more recent data points. This guarantees that even in cases when the lane boundaries are uneven or partially invisible, the identified lane center will remain visually steady.

4.3.3 Hybrid Kalman–EMA Optimization:

The research presents a hybrid optimization method that combines the stability of EMA smoothing with the predictive capacity of the Kalman Filter.

In this hybrid model:

- The Kalman Filter predicts the offset based on previous states.
- The EMA smooths out noise from sudden variations.
- Both outputs are combined to compute a hybrid offset, giving a more

accurate and consistent estimate.

Because lane markers on Indian highways are sometimes irregular or partially hidden, this hybrid technique provides strong real-time adaptability.

4.3.4 Performance Improvements:

The application of these optimization techniques results in:

- Increased lane detection consistency, achieving smoother line transitions.
- Improved offset stability, reducing pixel-level noise between consecutive frames.
- Higher frame rate (FPS) performance by efficient computation.
- Enhanced robustness under poor lighting, faded markings, or curved lanes.

The final output video shows dependable offset estimates and stable lane visualization, demonstrating the efficacy of these optimization techniques in real-time adaptive lane recognition for Indian roadways.

IMPORTANCE OF OPTIMIZATION IN LANE DETECTION:

In order to create a lane detecting system that is both precise and extremely dependable in real-world driving situations, optimization is essential. Lane detection demands the system to comprehend continuous lane structure, adjust to dynamic situations, and generate consistent outputs in real time. It goes beyond simply recognizing white or yellow lines on the road. Even the best detection algorithms can yield noisy, unstable, or delayed findings if they are not properly optimized.

A lane detection system needs to function reliably in a number of difficult situations:

1. Sharp Curves, Lane Splits, and Merges

Roads often include:

- tight curves,
- lane diverging sections (e.g., exits),
- merging lanes,
- S-shaped patterns,
- sudden widening or narrowing of lanes.

In such cases, the lane geometry changes abruptly, and the detection system must quickly adapt to new positions while still keeping the output smooth.
Unoptimized systems may fail to detect the new curve in time or may overshoot, causing unstable lane overlays.

Optimization ensures:

- faster curve adaptation,
- stable following of the lane trajectory,
- minimal lag during sudden geometry transitions.

2. Faded or Broken Lane Markings

Real-world roads often have:

- worn-out paint,
- partially visible lanes,
- broken dashed lines,
- construction zones with temporary markings.

These variations can confuse algorithms that rely purely on edge detection or color segmentation.

Optimization helps by:

- stabilizing output with predictive filtering (e.g., Kalman),
- maintaining lane continuity even when markings temporarily disappear,
- reducing the impact of missing edge segments.

This makes the system more resilient to poor road maintenance conditions.

3. Shadows, Glare, and Low-Light Conditions

Environmental conditions add unpredictability. A lane detection model must handle:

- harsh sunlight creating bright glare,
- shadows cast by trees, buildings, or vehicles,
- fog, dusk, or nighttime lighting,
- wet roads causing reflection.

In such cases, lane contrast reduces significantly, making it hard to detect edges.

Through optimization:

- thresholds adapt to lower visibility,
- smoothing filters help maintain lane stability despite noisy edges,
- predictive models compensate for temporarily masked lane portions.

Optimization ensures that detection quality remains high even in visually challenging scenarios.

4. High-Speed Driving and Rapid Lane Changes

At higher speeds, frame-to-frame lane positions shift quickly. The system must:

- track lanes with minimal delay,
- react instantly to changing road curvature,
- avoid jittering during rapid motion.

Without proper optimization:

- the output may lag behind the actual lane,
- real-time predictions will be inaccurate,
- sudden jumps in detection can occur, making the system unsafe for fast driving.

Optimization techniques ensure:

- real-time responsiveness,
- smooth transitions,
- stable tracking even when the vehicle is moving quickly.

5. Consequences of Poor Optimization

If optimization is not implemented correctly, a lane detection system may show:

a. Slow Response Time

The system becomes sluggish in updating lane positions due to:

- heavy computation,
- inefficient algorithms,
- improper tuning of filters.

This delay can be dangerous as the vehicle may drift before the system updates the lane boundaries.

b. Aggressive or Over-Sensitive Behavior

Conversely, if the system reacts too aggressively:

- lane lines appear shaky or jumpy,
- minor noise causes large movements,
- outputs become visually unstable.

This reduces the reliability and makes the lane detection appear inconsistent. In advanced driver-assistance systems (ADAS), such instability is unacceptable.

6. Optimization TThrough EMA Tuning

One of the key optimizations implemented is the fine-tuning of the Exponential Moving Average (EMA) smoothing factor.

Why EMA Matters?

EMA helps smooth the detected lane boundaries by averaging:

- current lane detection,
- past lane positions.

But its behavior depends on a parameter called the smoothing factor (α).

If α is too low (heavy smoothing):

- lane movement becomes very slow,
- system cannot adapt quickly to curves or lane merges,
- delays appear in detection.

If α is too high (light smoothing):

- lane detection becomes jittery,
- small fluctuations create noticeable instability,
- output becomes noisy.

Optimized EMA Finding

Through iterative tuning and testing:

- an optimal balance was found between responsiveness and smoothness,
- ensuring the system adapts quickly while staying visually stable.

This optimization significantly improves:

- lane continuity,
- reaction speed,

- overall real-time performance.

WHY EMA SMOOTHING?

The Exponential Moving Average (EMA) is essential in lane detection because it tracks lane boundaries smoothly, steadily, and consistently across video frames, the Exponential Moving Average (EMA) is crucial to lane recognition. Because to noise, shadows, glare, uneven illumination, worn-out lane markings, and partial occlusions, real-world lane recognition outputs frequently fluctuate, making the detected lane lines jitter or bounce from frame to frame if handled directly. In order to give more weight to new detections while maintaining the history of earlier positions, EMA combines the current lane position with previously smoothed readings. This weighted smoothing is particularly helpful on curves, merges, and roads with faded markers since it guarantees that the system does not overreact to minor perturbations and maintains a natural, progressive flow of lane movement.Because previous smoothed data minimize abrupt jumps, EMA also helps the system remain stable when lane lines momentarily vanish due to cars obstructing the camera or poorly visible road paint. The system strikes the ideal balance between responsiveness and stability by adjusting the smoothing factor (alpha); higher alpha values enable rapid adaptation to real changes in road geometry, while lower values offer more robust smoothing against noise. EMA is also computationally efficient, which makes it appropriate for high FPS real-time processing without adding complexity. All things considered, EMA smoothing significantly improves the lane detection pipeline's dependability, continuity, and visual stability, guaranteeing a more resilient performance under a variety of road and environmental

situations.

EFFECT OF ALPHA ON LANE STABILITY:

The stability or responsiveness of the lane detection output is largely dependent on the value of alpha (α) in the Exponential Moving Average (EMA). A low alpha, usually between 0.2 and 0.5, results in very reliable and smooth lane tracking by giving the prior smoothed lane positions a lot more weight. Even in difficult situations like shadows, reflections, or partially fading marks, this strong averaging lowers jitter, suppresses noise, and produces a serene, visually consistent lane overlay.The disadvantage is that a low alpha prioritizes the previous data over the present frame, which causes the system to react slowly to quick or sharp changes in road geometry, such as tight curves, lane mergers, or abrupt lane deviations. Conversely, a high alpha—typically between 0.8 and 1.0—makes the system far more responsive and quick to adapt to sudden changes in lane position. Sharp maneuvers and dynamic driving situations where lane boundaries alter quickly from frame to frame benefit from this great responsiveness. However, this also increases the system's sensitivity to noise from each frame, making it less stable. The recognized lane lines may jitter or fluctuate considerably due to little changes in edges, shadows, or camera vibrations.As a result, selecting the proper alpha value requires striking a balance between responsiveness and smoothness; if it is too high, the detection becomes unstable, and if it is too low, the system becomes slow. Optimal tuning guarantees that the lane recognition is both adaptable to actual road changes and stable in the face of noise.

- Low Alpha (0.2 - 0.5) :

  Produces smoother lane tracking by heavily averaging past detection's. Can be slow to react to sudden lane changes

(sharp curves)

- High Alpha (0.8 to 1.0):

  Fast response to lane changes but introduces instability in tracking. More sensitive to noisy detection's, leading to jittering.

BAYESIAN OPTIMIZATION FOR FINE-TUNING EMA

Bayesian Optimization is a technique used to find the best hyper-parameters by

systematically evaluating different values and improving predictions based on prior results. Instead of manually testing different alpha values, Bayesian Optimization selects and refines parameters automatically reducing trail-and-error efforts.

HOW IT WORKS?

STEP 1: DEFINE THE OBJECTIVE FUNCTION:

The system evaluated lane tracking performance based on smoothness stability and real- time adaptability.

STEP 2: BAYESIAN OPTIMIZATION BEGINS:

The algorithm starts by testing a few initial alpha values and measuring how well they perform.

STEP 3: ITER ATIVE LEARNING:

Using past results, Bayesian Optimization predicts better alpha values for the next iterations, converging towards the optimal value.

STEP 4: CONVERGENCE TO BEST ALPHA:

After several iterations, the system identifies the best alpha (0.64) which provides the best balance between lane stability and responsiveness.

4.4POST PROCESSING IN LANE DETECTION:

Post-processing is a crucial step in lane detection systems because it converts the raw, flawed output of the U-Net segmentation model into a clean, continuous, and dependable lane structure appropriate for real-world driving situations, post-processing is a crucial step in any lane recognition system. Even while U-Net offers pixel-wise lane segmentation, the predictions may still include distortions from lighting or shadows, jitter from frame to frame, thickness abnormalities, broken lane areas, and false positives. Because of these flaws, the lanes are dangerous and unstable for use in downstream functions like trajectory planning, lane departure warning, and steering control.

In order to improve the continuity and usability of the identified lanes, post-processing employs a series of refinement operations. Typical procedures include contour extraction to determine the most confident lane forms, polynomial fitting to transform raw segmented pixels into smooth, mathematically defined curves, and morphological operations (such as dilation and erosion) to fill in small gaps and smooth lane boundaries. Furthermore, methods like connected-component labeling, sliding-window search, and clustering aid in separating actual lane sections and eliminating noise or superfluous structures.

Post-processing becomes even more crucial in video-based lane detection since lane predictions need to hold steady over successive frames to avoid flickering or abrupt jumps. Therefore, lane curvature, position, and angle are stabilized using temporal smoothing techniques as Kalman filtering, exponential moving averages (EMA), or hybrid filters. Even when the model faces difficult frames with rain, shadows, worn-out markings, or occlusions, these techniques guarantee that the recognized lanes appear continuous and constant.

All things considered, post-processing serves as an intelligent correction layer that transforms the raw segmentation output into a refined, reliable, and accurate depiction of the lane geometry. It guarantees that the final visualized lanes have a clean, smooth appearance and offer accurate measurements, allowing the system to function well in actual driving situations.

4.5DATASET PREPARATION:

A custom data-set was created by extracting video frames from traditional lane detection outputs. Each frame was re-sized to 224 * 224 pixels to ensure uniformity in training. The dataset was then labeled to create a binary lane masks, where masked as white (1) and the background as black (0).

STEPS IN DATASET PREPARATION:

1. FRAME EXTRACTION: Captures images from lane detection videos.
2. REZISING: Standardize all images to 224 * 224 pixel for consistency .
3. LANE MASKING:Used HSV color threshold to isolate lane markings.
4. DATA AUGMENTAON: Applied flipping , rotation , brightness changes to improve model robustness.

This dateset was then used to train a U Net model for lane segmentation.

4.6DEEP LEARNING MODEL:

The U Net architecture was chosen for lane detection because of its ability to perform pixel-wise segmentation while preserving spatial details. The model consists of two main components :

ENCODER: Extracts lane-related features using convolutional layers.

DECODER: Reconsturcts lane masks from encoded features using upsampling layers.

4.6.1 TRAINING PROCESS:

INPUT: 224 * 224 images with normalized pixel values.
OUTPUT: Binary mask indicating lane positions.
LOSS FUNCTION: BCEWithLogitsLoss for pixel classification.
OPTIMIZER: Adam for efficient weight updates.
HARDWARE: Trained using GPU acceleration for faster convergence.

Once trained the model was integrated into the lane detection pipeline, where post- processing was applied to improve real-world performance.

4.7POST-PROCESSING TECHNIQUES FOR LANE STABILITY:

4.7.1 MORPHOLOGICAL FILTERING:

After obtaining the lane mask from the U Net model, morphological operations were applied to improve lane continuity. These operations helped in:

- Removing small noise pixel that were misclassified as lanes.
- Connecting broken lane markings by filling small gaps.

4.7.2 LANE OFFSET CALCULATION:

To determine the vehicle's position relative to the lane, the lane offset was calculated using:

Lane offset=lane center - vehicle center Lane center: Midpoint between

detected left and right lanes. Vehicle center: center of the camera frame.This calculation helps in assessing lane deviation, which is crucial for lane keeping systems.

4.7.3 STABILIZING LANE TRACKING WITH KALMANN FILTERING AND EMA:

Even after lane detection, slight variations across frames can cause instability. To smooth the tracking a Kalmann Filter + EMA hybrid approach was implemented.
Kalman Filter: Predicts lane positions in future frames, ensuring stable tracking even during temporary occlusions.
Exponential Moving Average (EMA): Reduces flickering by averaging lane positions over time, improving tracking stability.

4.8 PERFORMACE EVALUATION:

The system's performance evaluation demonstrates the significant influence of post-processing on lane detection accuracy and visual smoothness. The combination of Kalman + EMA hybrid filtering produced lane transitions that were substantially smoother, removing jitter and abrupt leaps that are common in raw frame-by-frame forecasts. Maintaining dependable lane monitoring, particularly in dynamic driving situations, depends on this steadiness.

Moreover, morphological treatments such gap filling, erosion, and dilation were crucial in lowering segmentation noise. Even in scenarios with fading, worn-out, or partially obscured lane markings, the system was able to preserve continuous and clearly defined lane structures thanks to these procedures, which successfully eliminated tiny artifacts and filled in broken lane segments.

Thanks to reliable lane offset computations, the system also showed excellent performance in vehicle positioning precision. The system could accurately calculate the vehicle's lateral deviation from the lane center by employing revised lane boundaries, enabling safer decision-making for autonomous or driver-assist applications.

It was evident from graphical analyses, such as frame-by-frame plots of lane curvature, offset stability, and detection consistency, that post-processing greatly improved lane dependability. These visual comparisons demonstrated increased temporal stability, smoother lane curves, and a significant decrease in detection errors, demonstrating that post-processing is necessary to achieve real-world-grade lane detection performance.

4.9 INTEGRATION OF TRADITIONAL AND DEEP LEARNING APPROACHES:

The system uses a well-balanced hybrid lane detection strategy that combines the advantages of deep learning-based segmentation, temporal post-processing, and conventional computer vision approaches to achieve both fast speed and good accuracy. Initially, quick and lightweight lane estimates are produced using conventional techniques like Canny edge detection and Hough Transform. These traditional techniques are perfect for rapidly detecting possible lane borders and offering a trustworthy first-stage approximation because they require little processing power.

Building on this basis, the system refines these first detections using a U-Net deep learning model. Under difficult real-world circumstances—such as shadows, dense traffic, poor visibility, curved lanes, and faded markings—where conventional approaches frequently fall short, U-Net excels at comprehending lane

topologies. The system attains a high degree of robustness in a variety of settings by merging classical and deep learning outputs.

Lastly, post-processing methods, particularly a hybrid Kalman Filter + Exponential Moving Average (EMA) arrangement, guarantee stability over time. While the EMA smoothes out abrupt oscillations, the Kalman filter forecasts lane movement across frames. When combined, they provide consistent, flicker-free lane tracking even in the presence of noisy raw detector data.

This integrated pipeline enables the system to achieve a high level of accuracy and reliability, which is necessary for advanced driver-assistance and autonomous driving applications, while also being computationally efficient, making real-time processing possible.

CHAPTER-5
TESTING

Test Cases for Evaluation

1. **Obtaining Video Streams**: Capabilities test the system's capability to successfully acquire video streams from uploaded video recordings. Ensure that the video feed taken matches the standards necessitated for lane detection and tracking.
2. **Pre processing Check**: Verify correct application of pre processing operations including re-sizing and edge detection on every video frame. Confirm whether color space transformation (HSV) and noise reduction processes such as Gaussian blur are applied to enhance the lane detection model accuracy.
3. **Lane Detection Using Edges and Color-Based Techniques**: Assess the robustness of the system in detecting lanes in various lighting, meteorological, and roadway conditions. The accuracy of lane localization should be verified: Lane markings in video frames must be correctly represented.
4. **DEEP LEARNING-BASED LANE DETECTION EVALUATION**: Evaluate

   the UNet model's lane detection accuracy using IoU. Verify lane consistency across frames,ensuring robustness against occlusions,faded lines and false positives.
5. **Offset Estimation with Kalman Filter and EMA:** Evaluate the effectiveness of the Kalman Filter in accurately estimating the car's offset from the center of the lane for contiguous frames. The effectiveness of the Exponential Moving Average in the reduction of lane boundaries and in maintaining an offset measurement.

6. **Lane Monitoring and Continuity**: The identified lane should also have been monitored without meaningful disruptions at time t consecutive frames. Test the tracking robustness for faded or broken lanes, or when other vehicles partially obscure the lanes.
7. **Post processing Verification:** Confirm the correctness of lane boundary predictions and make sure that only the items related to lane markings should not appear in the output. Confirm that your final product is good, and ready for the video where it will be demonstrated indicating continuous lane tracking for viewing.

CHAPTER-6

RESULTS

Comparision of results of previous models to our model under different conditions:

Fig 6.1 Condition: Broken Lanes

Fig 6.2 Condition: Shadows

Fig 6.3 Condition: Expansion Joints

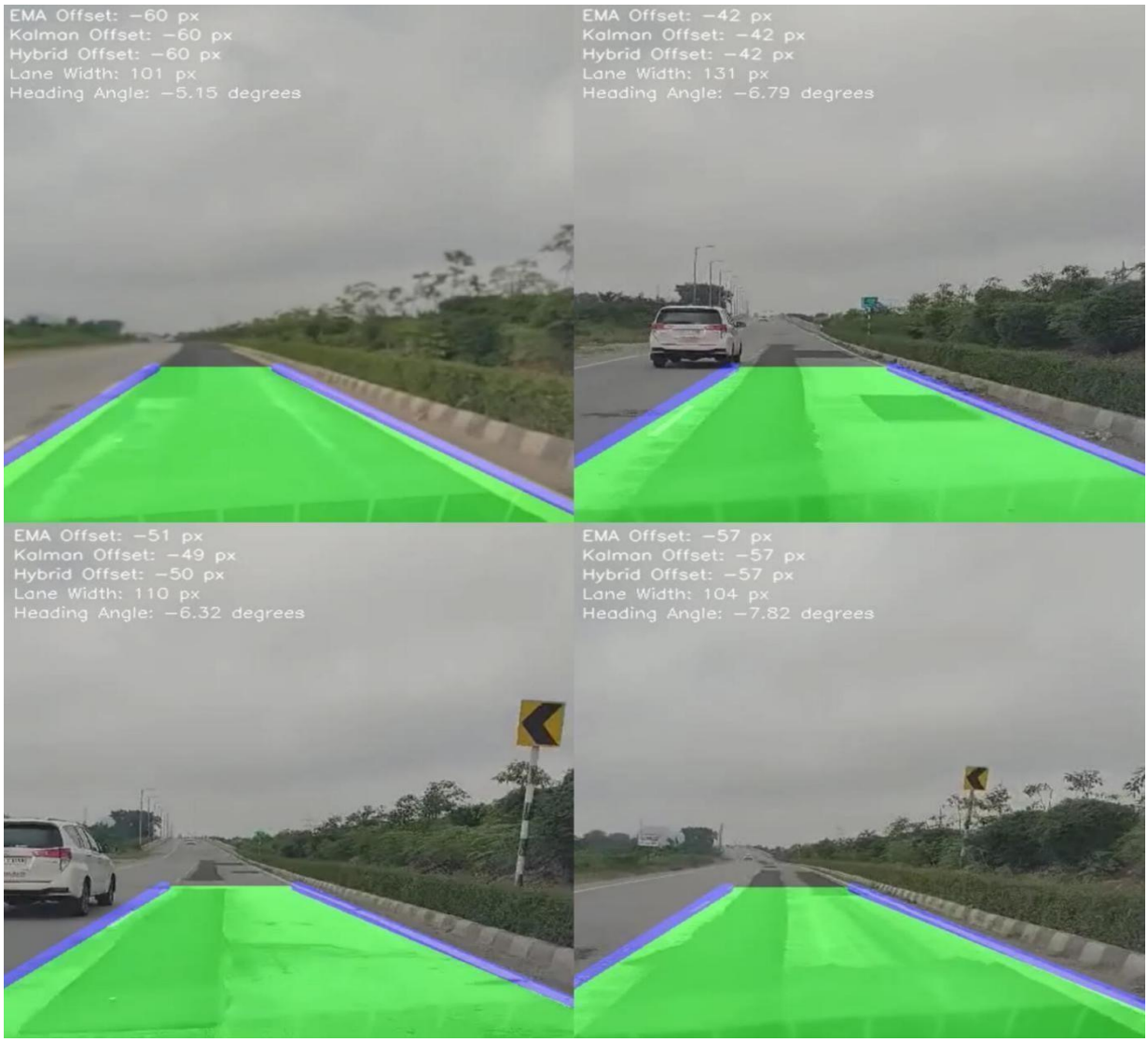


Fig 6.4 Output on Indian Roads

CHAPTER-7
CONCLUSION

The advanced lane detection system successfully designed has been able to successfully perform real-time identification of lanes and the tracking of a vehicle. The achievement was specially impressive considering the generally adverse road conditions, such as bad lane markings, the incidence of shadows, and the presence of expansion joints on the road. The proposed model, which provides a combination of conventional image processing methods with hybrid filtering techniques, is eligible to clearly detect the boundaries of lanes. In addition to this, it also makes accurate calculations of offsets so that the vehicle is exactly in the center of its allocated lane at any point of time in its journey.

This particular approach presents an exciting and motivational solution towards enhancing road safety and simultaneously facilitating the design and incorporation of autonomous driving technology. It is significant because vast and challenging road conditions are often found on Indian roads where general road environments can be extreme in many areas. Results obtained from this experiment show high detection and excellent real-time processing capabilities. These are very important results because they provide a basis from which further improvements can be engineered, especially targeted towards achieving enhanced accuracy in lane detection and overall stability in the systems. Some of the possible opportunities for interventions could be deep learning techniques that maximize the adaptability of such systems. The result would significantly contribute to the overall reliability of these intelligent driving technologies, ensuring better resilience in different conditions under which driving becomes operational.

CHAPTER-8
FUTURE SCOPE

The development of a Lane detection system utilizing OpenCV and other technologies offers numerous opportunities for future research and advancement:

- **Deep Learning Improved Lane Detection:** Techniques such as CNNs, which evolved into Transformer-based architectures in the newest deep learning architecture, could significantly improve the accuracy level of lane detection and allow following curvy roads, complex structures, and lanes that are poorly marked.
- **Improving Adaptability to Varying Road Surfaces:** This system would be supposed to improve on the functionality of lane detection responding to adverse conditions such as ill lit roads, extreme weather, heavy traffic, and others.
- **Real-time Performance Optimization:** It can be further optimized to cut down processing time and resource consumption to achieve flawless real-time lane detection for applications in autonomous driving and ADAS.
- **Dynamic Lane Change and Lane Obstacle Detection:** Developing the model for dynamic lane changes such as lane splits, exits, or merging lanes and detecting lane obstacles would make the system highly versatile and straightforward vehicle navigation.
- **ITS Integration:** V2I communication integration would therefore allow connectivity with the smart city infrastructure and, thereby, enable coordinated traffic management, adaptive routing, and automated response to lane usage and flow in traffic.
- **Collection of Data and Ongoing Learning**: Employ mechanisms of ongoing learning where the model is refreshed at regular intervals by new data taken from multiple different environments. This way, the model remains fresh and thus effective in different types of roads and geographies.
- **Real-time Feedback for Driver Assistance:** Even with real-time feedback,

lane deviation or potential lane departure warnings can easily support driver safety and enhance the current ADAS technologies provided in commercial vehicles. Exploring these future directions might make your lane detection project even significantly contribute toward the development of road safety.

CHAPTER-9
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